Lab 3: Architectural Pattern Selection – Written Justification

Project: Alumni Mentorship & Mock Interview Platform

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SECTION AIML-D

Architecture Selection

We chose Layered Architecture for the Alumni Mentorship & Mock Interview Platform.

Architectural Choice

We selected a Layered Architecture, organized into three layers as shown in the component diagram. The Client Layer (Web & Mobile UI) contains Student UI, Mentor Portal, and Admin Console. The Application Services Layer (Business Logic) contains the Mentor Matching Engine, Session & Booking Manager, Feedback & Scorecard Service, and Audit & Notification Service. The Persistence & External Services Layer contains the User & Profile DB, Session & Booking DB, and Calendar & Video Gateway. The layers communicate through named interfaces such as Matching API, Booking API, Feedback API, Profile Data Access, Session Data Access, and External Gateway.

Reason 1: Clear Separation of Concerns

The platform serves three main user groups — Students, Alumni Mentors, and Admins — with different responsibilities. Students search for mentors and book sessions, mentors manage availability and participate in mentorship or mock interviews, and administrators manage platform activities and feedback. A Layered Architecture separates these responsibilities clearly. The Client Layer handles user interaction, the Application Services Layer manages matching, booking, feedback, and notifications, while the Persistence & External Services Layer handles data storage and external integrations. This makes the system easier to understand, develop, maintain, and extend.

Reason 2: Well-Defined Workflow and Integration

The platform follows a predictable workflow in which users interact with the Client Layer, requests are processed by the Application Services Layer, and required data or external services are accessed through the lower layer. For example, a student can search for a mentor through the Mentor Matching Engine and book a session through the Session & Booking Manager. The Calendar & Video Gateway can then provide external scheduling and video-session integration without exposing these external services directly to the user interface. This structure makes it easier to modify or replace external services without significantly affecting the core business logic.

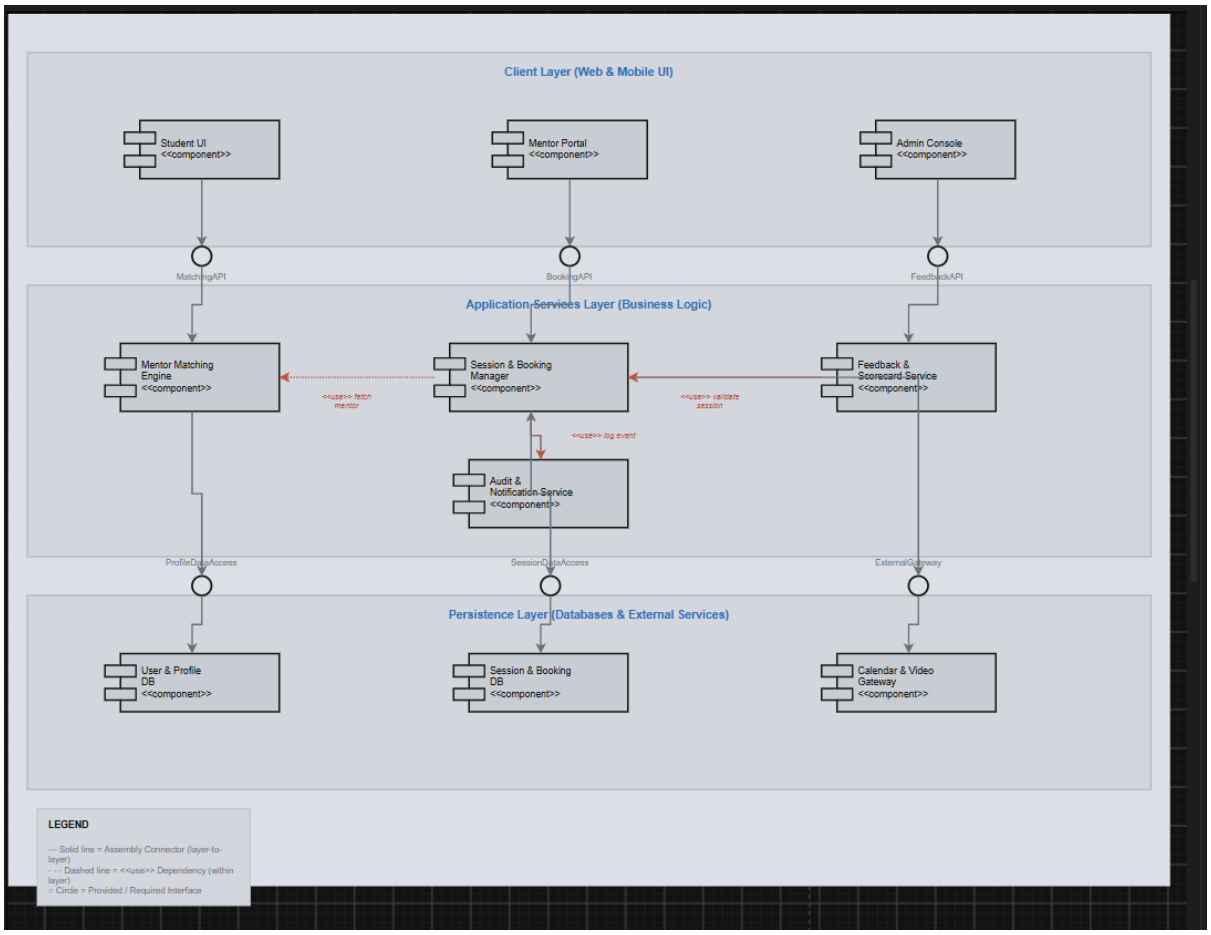
Security Advantage

The layered architecture provides controlled access to sensitive user and mentorship data. The Client Layer does not directly access the databases. Requests must pass through the Application Services Layer, where authentication, authorization, validation, and business rules can be applied before data is accessed. This helps protect information such as user profiles, mentor details, session records, and mock interview scorecards. Therefore, a problem in the user interface does not provide a direct path to the underlying databases.

Performance Benefit

The layered structure allows frequently used operations such as mentor search, availability checking, and session booking to be handled efficiently within the Application Services Layer. Frequently accessed mentor and availability information can be cached at this layer to reduce unnecessary database requests. Separating database operations from the presentation layer also prevents direct and unnecessary database access, helping the platform provide responsive interactions for students and mentors.

Component Diagram



Component Diagram Description

The component diagram represents the architecture of the **Alumni Mentorship & Mock Interview Platform** using the main components distributed across the Client, Application Services, and Persistence & External Services layers.

- **Student UI Component:** Provides the interface through which students search for alumni mentors, view mentor availability, book mentorship sessions, and access mock interview services.
- **Mentor Portal Component:** Provides the interface through which alumni mentors manage their availability, participate in mentorship sessions, and provide feedback or mock interview scorecards.
- **Admin Console Component:** Provides administrative functionality for managing platform activities, monitoring feedback, and maintaining the overall mentorship platform.
- **Mentor Matching Engine Component:** Handles mentor discovery and matching between students and suitable alumni mentors based on the platform's matching criteria.

- **Session & Booking Manager Component:** Manages mentorship session scheduling and booking operations, including checking availability and coordinating session details.
- **Feedback & Scorecard Service Component:** Handles feedback collection and post-interview scorecards submitted after mentorship or mock interview sessions.
- **Audit & Notification Service Component:** Records important system events and manages notifications related to mentorship sessions, bookings, and other platform activities.
- **User & Profile DB Component:** Stores and retrieves user profile information for students and alumni mentors required by the platform.
- **Session & Booking DB Component:** Stores and retrieves session, booking, and scheduling information used by the Session & Booking Manager.
- **Calendar & Video Gateway Component:** Provides integration with external calendar and video services required for scheduling and conducting online mentorship sessions.